ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications [ANSWER KEY]

Unit 6.01: Topic 6: AI SL Modeling, Financial Math & Chi-Squared Tests

Q1. If PA=0.3 and PB=0.5 and A,B are independent, find PAcup B.

[x] A: 0.65
[ ] B: 0.80
[ ] C: 0.15
[ ] D: 0.85

Solution: PAcup B=0.3+0.5-0.3times0.5=0.65 [M1] [A1]

Q2. The following bar chart shows the number of students who achieved different grades in a mathematics test.nThe grades are A, B, C, and D. The data is as follows: Grade A: 14 students, Grade B: 22 students, Grade C: 18 students, Grade D: 26 students.nWhat is the total number of students who took the test?

[ ] A: 70
[x] B: 80
[ ] C: 78
[ ] D: 60

Solution: To find the total number of students who took the test, we need to sum the number of students who achieved each grade.nNumber of students with Grade A = 14nNumber of students with Grade B = 22nNumber of students with Grade C = 18nNumber of students with Grade D = 26nTotal number of students = 14 + 22 + 18 + 26 = 80.nTherefore, 80 students took the test. [M1] [A1]

Q3. The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students surveyed?

[ ] A: 50
[ ] B: 60
[ ] C: 70
[x] D: 80

Solution: The bar chart shows the following counts for each sport: Sport 1: 14 students Sport 2: 22 students Sport 3: 18 students Sport 4: 26 students To find the total number of students surveyed, we sum the number of students for each sport: 14 + 22 + 18 + 26. 14 + 22 = 36 18 + 26 = 44 36 + 44 = 80. Therefore, the total number of students surveyed is 80. [M1] [A1]

Q4. The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

[x] A: 20
[ ] B: 25
[ ] C: 30
[ ] D: 35

Solution: The histogram displays the distribution of study hours for 100 students. The total number of students is 100. The question states that the number of students who studied for between 0 and 5 hours is 30. The bar for the '0-5 hours' interval has a height of 30, representing 30 students. The bars for the '5-10 hours' and '10-15 hours' intervals have heights of 20 and 15 respectively, representing 20 and 15 students. Therefore, the number of students who studied for between 5 and 10 hours is 20. [M1] [A1]

Unit 6.01: Topic 6: AI SL Modeling, Financial Math & Chi-Squared Tests

Q5. The bar chart below shows the number of students who achieved a certain score on a test. What is the total number of students who scored 6 or 8?

[x] A: 30
[ ] B: 28
[ ] C: 26
[ ] D: 32

Solution: From the bar chart, the number of students who scored 6 is 18 and the number of students who scored 8 is 12. The total number of students who scored 6 or 8 is the sum of these two values: 18 + 12 = 30. [M1] [A1]

Q6. The following two-way table shows the results of a survey on favourite sports among a group of students. Calculate the total number of students surveyed.

[ ] A: 50
[ ] B: 45
[x] C: 60
[ ] D: 55

Solution: The total number of students surveyed is the sum of all entries in the two-way table. text{Total} = 15 + 10 + 20 + 15 = 60 . [M1] [A1]

Q7. Find the mean of the data set: 20, 13, 12, 12, 8 .

[ ] A: 10
[x] B: 13
[ ] C: 8
[ ] D: 11

Solution: Working through this statistics step: Mean =dfrac{20+13+12+12+8}{5}=13 . [M1] [A1]

Q8. Find the mean of the data set: 10, 1, 5, 14, 3, 7, 0, 8 .

[x] A: 6
[ ] B: 4
[ ] C: 3
[ ] D: 1

Solution: Working through this statistics step: Mean =dfrac{10+1+5+14+3+7+0+8}{8}=6 . [M1] [A1]